PyGameMaker — Tutorial 4: Breakout — Teacher Guide

Companion for the in-app tutorial (**Help > Tutorials > Breakout: Brick Breaker**, 7 pages) and the matching student handout and worksheet. Students should have finished Tutorials 2 and 3 (events, collisions, score, draw, solid objects).

Overview

Students build Breakout in four phases and test after each: a bouncing ball, one row of bricks, four rows, and a game controller with lives. New ideas: **parent and child objects** (one collision rule for many brick types), a **death zone** instead of a bottom wall, **lives**, and a **Game Over with highscore**.

> The key idea is inheritance: "a brick red *is a* brick, so it gets whatever a brick gets". Ask students to predict what happens when the parent is not set on the new colour.

Suggested Timing (60-90 minutes)

The tutorial says 25-30 minutes for a confident student. It has seven pages and many objects, so plan a double period or two sessions.

Segment	Time	What happens
Recap and predict	5 min	Show the finished game; ask "which objects do we need?"
Sprites (page 2)	10 min	Let students import the sample images from Tutorials/04_breakout/assets/ to save time
Paddle, ball and room (pages 3-4)	20 min	Phase 1; everyone plays a bouncing ball
Bricks (pages 5-6)	20 min	Phases 2-3; the parent object
Controller, lives, Game Over (page 7)	15 min	Phase 4
Worksheet / exit ticket	5-10 min	Parts A-C

For a 45-minute slot, do sprites and Phase 1 in the first session and the rest in a second.

Step-by-Step Walkthrough and Common Issues

Phase 1: bouncing ball

- Two wall objects (side and top) exist so the ball can reverse the right direction: side wall reverses horizontal, top wall and paddle reverse vertical.
- **Solid** on walls and paddle is needed; the ball is *not* Solid. Every bounce is a separate Collision event on the ball.
- The paddle needs its own **Collision with obj_wall_side** event with *Stop movement*; otherwise it goes through the side walls.

- **There is no bottom wall on purpose.** The invisible death zone at the bottom catches the ball and jumps it to (320, 100). Students often place a wall at the bottom; ask "how would the ball ever get lost?".
- The ball keeps bouncing forever between walls and the paddle; that is **expected**. The paddle is only needed to stop the ball from falling.

Phase 2-3: bricks and the parent object

- obj_brick_parent has no sprite and is Solid. Each colour needs **Parent = obj_brick_parent** and **Solid**. Forgetting the parent on a new colour is the classic mistake: that colour is a wall that never breaks.
- The ball's brick event is on obj_brick_parent only. Show students that adding a fifth colour needs **no** change to the ball.
- Each brick gives 10 points regardless of colour (the tutorial leaves different scores as a challenge).
- Use **Snap to Grid** (32×16) when placing bricks, or rows will be uneven.

Phase 4: game controller

- The controller must be **placed in the room**, like the controller in Tutorial 2.
- Lives are set in the controller's **Create** event; the ball subtracts a life in its death-zone event (*Add to lives -1* before the jump).
- **No More Lives** is an event that fires automatically when lives reach zero. The tutorial asks for *Show message*, *Show Highscore*, *End Game* in that order and warns that order matters. On the desktop player the highscore is shown either way, but keep the tutorial's order: it is the safe one.
- The score and lives are drawn in white at y 10, over the grey top wall row; they stay readable.

> **Reference projects.** Download the checkpoint projects from the wiki (04_breakout_checkpoints.zip): one project for the end of each phase. Give a stuck student the previous phase so they can continue.

Vocabulary Introduced

Term	What it means here
Parent object	A template; children inherit its events
Child object	An object whose Parent field names another object
Death zone	An invisible object at the bottom that catches the ball
Lives	A counter of how many times the ball may be lost
Highscore table	A saved list of the best scores

Discussion Questions

- Why write one collision event for the parent instead of one per colour?
- Why does the ball reverse *vertical* movement on a brick?
- What could we change so the game is harder or fairer? (Speed, paddle size, angle of bounce.)
- Which parts of the game are events, and which are actions?

Differentiation

- **Support:** give the Phase 1 checkpoint and start at bricks; import the sample sprites instead of drawing.
- **Extension:** different points per colour; ball speeds up; a second level with another brick layout; a sound at the end.

Worksheet Answer Key

Part A: 1-C, 2-A, 3-B, 4-D.

Part B: 1. Reverse horizontal movement. 2. Destroy other instance. 3. Add to lives -1 (set lives, relative -1). 4. Show message. 5. End Game.

Part C:

1. Set its **Parent** to obj_brick_parent (and Solid). No, the ball's event for the parent already covers it.
2. Up and to the right. On the screen y grows downward, so a negative vertical speed means up.
3. Show Highscore must come before End Game because the tutorial says ending the game first can stop the highscore from appearing. (In the desktop player both orders work; the safe order is still best.)

Parts D and E: completion and reflection.

Rubric: Breakout

Level	What the game shows
4 - Complete	Paddle and ball work; bricks of all colours break through one parent event and give points; lives and score are displayed; Game Over, highscore and end work
3 - Working	Paddle, ball and bricks work; lives or Game Over is missing
2 - Partly there	Paddle and ball work; bricks do not break, or only one colour breaks
1 - Started	Objects exist, but the ball does not bounce or the paddle does not move

End-of-Session Checklist

- [ ] Every student has a saved project (Ctrl+S)
- [ ] Every student reached at least Phase 2 (a brick can be broken)
- [ ] Note who needs the checkpoint project next time